Model Trade-offs

In lending, predicting Loan_Status presents a classic trade-off between **risk** (approving bad loans) and **revenue** (approving good loans).

- **Logistic Regression (Winner):** This model offers the most balanced performance and highest ROC-AUC. By implementing class_weight='balanced', the model penalizes misclassifying the minority class (Rejected/Bad Loans) more heavily.
  - *Business Impact:* It successfully identifies and rejects **~71% of actual bad loans**. While this drops the approval rate of *good* loans slightly (from 99% to 88%), it results in a 31% reduction in total defaults compared to a baseline model.
- **Random Forest:** While it has a slightly higher recall for good loans (approving more applicants), it is weaker at distinguishing the underlying classes overall (lower ROC-AUC) and lets more default-prone loans slip through.

Suggested Deployment Threshold

By default, models use a 0.50 probability threshold to decide between Approved and Rejected. However, for deployment, you should adjust this threshold based on the institution's current economic strategy:

1. **Risk-Averse Strategy (Economic Downturn):** Increase the decision threshold to **0.60 or 0.65**. The model will only approve loans it is highly confident in. This will lower your overall approval rate and sacrifice some interest revenue, but strictly protect capital from defaults.
2. **Market-Share Strategy (Growth Phase):** Lower the threshold slightly to **0.49** (the statistically optimal point on the ROC curve, via Youden's J statistic). This captures the maximum number of good borrowers while still respecting the mathematical boundaries of the risk profile.